

Neeraj Jadhav

Education

- 2024-26 **Master of Science, Computer Science**, Specialization: Computing Systems
Master's Thesis: Can LLMs do program behavior analysis and optimization?
Prof. Santosh Pande, Georgia Institute of Technology, GPA: 4.0
- 2020-24 **Bachelor of Technology with Honors, Computer Science and Engineering**
Minor in Artificial Intelligence and Data Science
Indian Institute of Technology, Bombay, (Major) CGPA: 9.34, (Minor) CGPA: 9
Key courses: Algorithms, Operating Systems, Compilers

Publications

- **Phaedrus: Exploring Dynamic Application Behavior with Lightweight Generative Models and Large-Language Models** (Submitted to PLDI, 2025)
Bodhisatwa Chatterjee, Neeraj Jadhav, Sharjeel Khan, and Santosh Pande

Honors & Awards

- Recipient of **Masters Graduate Fellowship in CS** by *Georgia Institute of Technology* ['24]
- Secured **46th/150k** and **186th/870k** national ranks in **JEE Advanced** and **Main** exams resp. ['20]

Professional Experience

- Summer '23 **Quantitative Research Internship, Quadeye Securities LLP**, Gurugram, India
○ Engaged in **large-scale** global optimization of **trading strategies** using **Bayesian** Optimization, **Genetic** & **Whale** Algorithms achieving **80% reduction** in **time** spent and **2% increased PnL**
- Winter '22 **Software Engineering Internship, FinIQ Consulting**, Pune, India
○ **Enhanced** the proprietary financial coding language **FinSPL**, by adding **operational parallelism** and **efficient** data structures for information **storage** and **retrieval** of streaming **market indices**
- Summer '22 **Automated Orchestration using LLMs, IBM Research**, Gurgaon, India
○ **Streamlined** business task execution using **LLMs** and **process mining** to optimize **workflow efficiency**
○ Pipelined **data extraction** from **conversation logs** by fine-tuning **GPT2** for seamless **API calls**

Research Experience

- Fall '23 **Robust Celltree : A Paradigm for Distributed Data Repositories**, *Bachelor's Thesis*
Prof. Manoj Prabhakaran (IIT Bombay), Prof. Indranil Gupta (UIUC)
○ Designed a **new data structure** with **safety** guarantees to **minimize** computational **load** and **time**
○ Enhanced **blockchain** with **proof-of-stake** and modified **consensus**, achieving high **latency gains**
- Fall '23 **mpC: A Secure Multiparty Computation Language**
Prof. Manoj Prabhakaran (IIT Bombay)
○ Developed a **language**, with backend in **C++**, for **abstracting** out **cryptographic protocols**
○ Secured **communication with TLS** & **reduced** compilation **time** for **rapid testing** of new protocols

Key Projects

- Summer '24 **Trading Simulator and Backtester**, Personal Project
- Spring '23 **ipIC : Miniature GCC-like C Compiler**, Guide: *Prof. Amitabha Sanyal (IITB)*
- Spring '23 **Blockchain Simulator**, Guide: *Prof. Vinay Ribeiro (IITB)*
- Spring '22 **Peer-to-Peer Networked File Sharing Application**, Guide: *Prof. Kameswari Chebrolu (IITB)*
- Spring '22 **RISC 16 Bit Processor in VHDL**, Guide: *Prof. Virendra Singh (IITB)*

Leadership

- Mentorships **Student Mentor & Tech Mentor at XLR8 Robotics competition, IIT Bombay**
- TAships **Calculus & Embedded Systems, IIT Bombay, Algorithms, GATech**

Technical Skills

- Languages & Libraries** C/C++, Python, R, JavaScript, SQL, Bash, Huggingface, PyTorch
- Software Frameworks** Git, MATLAB, Wireshark, Django, React.js, Node.js

Extracurricular Activities

- Member of the **Campus Cats Club** at GATech, supporting **initiatives to care for** campus cats
- Amongst **top 10** teams in **Algorithmic Trading** contest organized by Web and Coding Club, IITB
- Engage in playing the **violin** to enhance musical proficiency, contributing to theatrical **performances**